Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 12 · SLO 1 — Solved Worksheet

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Unit â€“ 3 : Session â€“ 12 : SLO â€“ 1 | Circular Queue â€” Dequeue Operation

■ I. Algorithm to Delete an Element from a Circular Queue

• Step 1: IF FRONT = -1
• Write UNDERFLOW
• Goto Step 4
• [END OF IF]
• Step 2: SET VAL = QUEUE[FRONT]
• Step 3: IF FRONT = REAR
• SET FRONT = REAR = -1
• ELSE
• IF FRONT = MAX-1
• SET FRONT = 0
• ELSE
• SET FRONT = FRONT + 1
• [END OF IF]
• [END OF IF]
• Step 4: EXIT

■ II. Programming Code â€” delete_element()

• int delete_element() {
• int val;
• if (front == -1) {
• printf("\n UNDERFLOW");
• return -1;
• }
• val = queue[front];
• if (front == rear)
• front = rear = -1;
• else {
• if (front == MAX-1)
• front = 0;
• else
• front++;
• }
• return val;
• }